

# A Research Principle for the AI Era

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**Even when intrinsic value is the sole objective, the optimal strategy starts by investing entirely in external value.**

The model below establishes this result for a sufficiently long research horizon.

Intrinsic value is the scientific contribution the researcher ultimately values. External value is support convertible into research funding, measured in the same units as expenditure.

## Goal

Maximize expected total intrinsic value over  $n$  research cycles:

$$\max \mathbb{E} \left[ \sum_{i=1}^n I_i \right].$$

## Assumptions

Consider  $n$  research cycles, starting with a seed budget  $c_1 = 1$ . In cycle  $i$ , expenditure is  $c_i$  and the fraction allocated to intrinsic work is  $p_i \in [0, 1]$ .

**1. Linear returns.** Intrinsic value, external value, and the next budget satisfy

$$I_i = a_i p_i c_i, \quad E_i = b_i (1 - p_i) c_i, \quad c_{i+1} = d_i E_i.$$

The vectors  $(a_i, b_i, d_i)$  are iid across cycles. All coefficients are bounded above, and  $a_i$  has a positive lower bound. Let  $b$  be the lowest possible value of  $b_i$ . Assume  $d_i \geq 1$ : external support generates at least an equal research budget in the next cycle.

**2. A growing reserve.** The funding balance obeys

$$B_0 = 0, \quad B_i = B_{i-1} + E_i - c_i.$$

For every possible coefficient outcome, require

$$B_i \geq \rho B_{i-1}, \quad 1 < \rho < b.$$

Research income must cover spending while growing the reserve.

**3. Decision information.** The researcher observes the current budget and balance and chooses  $p_i$  before the current coefficients are drawn. Only  $b$ ,  $\mathbb{E}[b_i]$ , and  $\mathbb{E}[b_i d_i]$  are known about the common distribution.

## The result

**Theorem 1.** *There is a finite integer  $K \geq 1$ , independent of the horizon  $n$ , such that whenever  $n > K$ , the unique optimal decision in each of the first  $n - K$  cycles is to allocate all resources to external value:*

$$p_i = 0 \quad (1 \leq i \leq n - K).$$

## A numerical example

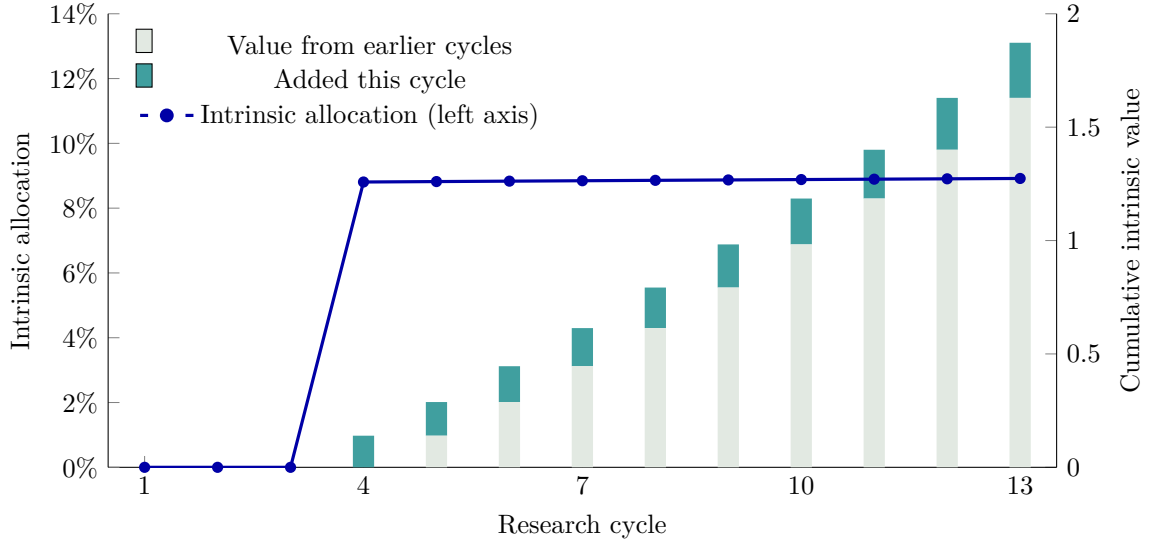


Figure 1: An optimal allocation: the first three cycles produce no intrinsic value because all resources go to external work. The line shows the intrinsic allocation share. Each bar shows cumulative intrinsic value, with the current increment highlighted.

Take  $n = 13$ ,  $a_i = 1$ ,  $b_i = 1.11$ , and  $d_i = \rho = 1.05$  in every cycle. Here  $K = 10$ : maximizing total intrinsic value begins with three cycles devoted entirely to external value.

## Proof

Every feasible state satisfies  $0 \leq B \leq c$ : this holds initially and is preserved because

$$0 \leq B' = B + E - c \leq E \leq d_i E = c'.$$

Write  $u = (1 - p)c$  for external expenditure. The reserve constraint gives the feasible interval

$$\frac{c + (\rho - 1)B}{b} \leq u \leq c.$$

Independence lets us divide the objective by the positive constant  $\mathbb{E}[a_i]$ , giving stage reward  $c - u$ . Set

$$w = \mathbb{E}[b_i], \quad v = \mathbb{E}[b_i d_i], \quad v \geq w \geq b > \rho > 1.$$

The value function with  $m$  cycles remaining is linear:  $V_m(c, B) = A_m c + D_m B$ , with  $A_0 = D_0 = 0$ . To see this, define  $S_m = vA_m + wD_m$ . The Bellman objective is

$$(1 - D_{m-1})c + D_{m-1}B + (S_{m-1} - 1)u.$$

Its maximizing endpoint depends only on  $S_{m-1}$ , and both endpoints are linear in  $(c, B)$ , completing the induction.

When  $S_{m-1} \leq 1$ , choose the lower endpoint. Starting from  $S_0 = D_0 = 0$ , this gives

$$D_m = D_{m-1} + \frac{(\rho - 1)(S_{m-1} - 1)}{b} \leq 0$$

and

$$\begin{aligned} S_m &= \frac{v + w(\rho - 1)}{b} S_{m-1} + \frac{v(b - 1) - w(\rho - 1)}{b} \\ &\quad + (w - v)D_{m-1} \geq \rho S_{m-1} + b - \rho. \end{aligned}$$

Since  $\rho > 1$  and  $b > \rho$ , this sequence crosses 1 in finite time. Let  $K = \min\{m \geq 1 : S_m > 1\}$ .

Once  $S_{m-1} > 1$ , the unique maximizing choice is  $u = c$ . Then  $D_m = D_{m-1} \leq 0$  and

$$S_m = vS_{m-1} + (w - v)D_{m-1} > S_{m-1} > 1.$$

Thus every decision with more than  $K$  cycles remaining uniquely allocates everything to external value. These are precisely the first  $n - K$  cycles.  $\square$

**Even when only intrinsic value matters, the optimal starting allocation is entirely external.**